Regular maps for the O'Nan group: a corrected source synthesis

Human Compression project

Abstract

We assemble the acquired proofs and finite certificates for Connor and Leemans, *Algorithmic enumeration of regular maps*. The corrected triangular table has 136 entries and sum 798644. Its positive types number 129; the chiral, inner-reflexible and outer-reflexible type spectra have sizes 128, 95 and 97. We state the counting convention, the imported group and character data, the source corrections, and the separate evidence for each of the source's mathematical clauses. The construction and verification of witnesses are distinguished from exhaustion of markings. In particular, we do not claim the total numbers of inner- or outer-reflexible markings. This manuscript joins already acquired components; it does not claim a new classification of a sporadic group or a new general enumeration theorem.

1 Objects, conventions and the corrected result

Let $G = \mathrm{O'N}$ and $A = \mathrm{Aut}(G) = G : 2$. For $p, q > 2$ put

$$E_G(p,q) = \{(R,S) \in G^2 : |R| = p,\ |S| = q,\ |RS| = 2, \langle R, S\rangle = G\}, \qquad n(p,q) = |E_G(p,q)/A|.$$

The action is simultaneous application of an automorphism to both entries. Pairs are ordered. Displaying only $p \le q$ chooses one orientation of different-order types; when $p = q$, no additional division by duality is performed. The source sometimes writes braces around the pair. Throughout this treatment its numerical convention is the ordered-pair convention of its equation (2.4). A quotient identifying dual markings is a different problem.

A marking is reflexible if an automorphism simultaneously inverts R, S. It is inner-reflexible or outer-reflexible according as this automorphism lies in G or $A \setminus G$; otherwise it is chiral. These alternatives are disjoint. Their map and polyhedron interpretation uses the correspondence proved in Section 4 and the classical map interpretation retained from the source.

Write

$$P = \{3,4,5,6,7,8,10,11,12,14,15,16,19,20,28,31\},$$
$$Z = \{(3,3),(3,4),(3,5),(3,6),(3,7),(3,12),(4,4)\}.$$

and $T = \{(p,q) \in P^2 : p \le q\} \setminus Z$.

Theorem 1.1. *Under the named inputs in Section 2, the acquired component proofs and certificates establish the following corrected conclusions.*

1. *The exact values of $n(p,q)$ are Tables 1–2. Their sum is* 798644*; their positive support is T, of size* 129*.*
2. *Chiral markings occur precisely at $T \setminus \{(3,15)\}$, of size* 128*.*
3. *Inner-reflexible markings occur precisely at*

$$I = \{(p,q) \in T : 20, 31 \notin \{p,q\}\} \setminus \{(3,10),(4,5),(4,6)\}, \qquad |I| = 95.$$

4. Outer-reflexible markings occur precisely at

$$J = \{(p,q) \in T : 8,16 \notin \{p,q\}\} \setminus \{(6,12)\}, \qquad |J| = 97.$$

The table counts automorphism orbits of markings. The spectrum sizes $129, 128, 95, 97$ count *types*. No total of all inner-reflexible or all outer-reflexible markings is inferred from these type counts. The proof of the theorem is the following synthesis of the explicitly listed component proofs, not a new replay of every numerical calculation.

2 Named inputs and their provenance

The following are explicit imported mathematical data, rather than facts deduced merely by matching a group order or a program label.

1. The identification of the O'Nan simple group, its order $g = 460815505920$, its centrelessness, its full automorphism group $G : 2$, and its nonidentity element-order set $P \cup \{2\}$ are retained Atlas inputs. The acquired degree-245520 permutation representation of $\mathrm{O'N} : 2$ is faithful. Its named socle has an orbit of size 122760. The literal permutation readers check words and equalities in this model; they do not independently classify the named group.

2. The complete subgroup-class roster, normalizer orders, maximal classes and the specifically used direct inclusion multiplicities are retained from Connor–Leemans' subgroup-lattice atlas. There are thirteen maximal classes, of orders

   $$3753792, 3753792, 175560, 161280, 25920, 25920,$$
   $$14880, 14880, 10752, 7920, 7920, 2520, 2520.$$

   The full roster has 581 subgroup classes. Unweighted containment alone does not supply arbitrary weighted incidences. Each component requiring an actual inclusion multiplicity proves or acquires that multiplicity; it does not multiply weights along arbitrary cover paths.

3. The ordinary character tables named `ON`, `ON.2` and the subgroup tables used in the numerical components are the acquired Atlas/CTblLib data. Their stored class fusions, order maps, power maps and identifications are retained inputs. Character arithmetic and class joins are checked exactly. Column norms and order consistency supplement these inputs; they are not replacements for their named identifications.

4. The complete centralizer, pair and word-transport certificates used for involution enumeration are imported at their precise acquired scope. In particular $|C_G(z)| = 161280$ for an inner involution z, $|C_A(z)| = 322560$, and an outer involution t has $C_A(t) = \langle t \rangle \times J_1$. The acquired commuting domains have 1750 inner involutions other than z, and 1463 outer involutions other than t. Small-action faithfulness and full stabilizers are separately checked before their orbit counts are used.

The classical Frobenius character identity, Hall subgroup inversion and the map interpretation retain their original attribution. Woldar's non-Hurwitz theorem supplies the cited exclusion $(3, 7)$; our use of that theorem does not constitute a recovery of Woldar's entire paper. The Conder–Oliveros attribution for the simple-group polyhedron interface is retained, together with the independent intersection-property proof below. Source reports of old running times and earlier authors' results remain historical attributions.

The companion relative-path binding list identifies the exact retained source, model, table, proof and review interfaces. An acceptance record certifies that the indicated component was checked; it is not substituted for the component's mathematical proof or for its stated borrowed inputs.

3 Counting pairs without overcounting subgroups

For a finite group H, let $X_H(p,q)$ be the set defined as $E_H(p,q)$ without the generation condition, and write $x_H = |X_H|$, $e_H = |E_H|$. For a conjugacy-class representative c_i and an irreducible character χ, put $v_{\chi,m} = \sum_{|c_i|=m} |c_i^H| \chi(c_i)$. Frobenius' formula gives

$$x_H(p,q) = \frac{1}{|H|} \sum_{\chi \in \mathrm{Irr}(H)} \frac{v_{\chi,2} v_{\chi,p} v_{\chi,q}}{\chi(1)}. \tag{1}$$

Indeed $(R,S) \mapsto ((RS)^{-1}, R, S)$ is a bijection to the order-$(2,p,q)$ triples with product one. The sum over their conjugacy classes is exactly (1). All arithmetic is in the exact character field, with the final nonnegative integer checked exactly.

Lemma 3.1. *For every finite H,*

$$e_H(p,q) = x_H(p,q) - \sum_{K<H} e_K(p,q).$$

Grouping actual subgroups by H-conjugacy replaces each term by $[H : N_H(K)]e_K(p,q)$. The action of $\mathrm{Aut}(H)$ on $E_H(p,q)$ is free.

Proof. Every pair in X_H generates one unique subgroup, so the sets E_K for actual $K \le H$ partition X_H. Conjugate subgroups have equal intrinsic counts and their number is the normalizer index. An automorphism fixing both generators fixes every word in them, hence the whole group. □

Consequently the correct denominator for G is $|A| = 2g$, applied to generating pairs, not to all character-counted pairs. In particular,

$$n(p,q) = \frac{1}{2g} \left(x_G(p,q) \; - \; \#\{\text{nongenerating pairs in } X_G(p,q)\} \right).$$

One may use a complete weighted subgroup matrix and solve a unit triangular system in increasing subgroup order. The lcm filter is valid because a subgroup containing elements of orders p, q has order divisible by $\mathrm{lcm}(p,q)$. This is the source's finite dynamic program and Hall inversion, not a new inversion theorem.

For this application the numerical components instead exploit exact maximal-overgroup and generated-subgroup decompositions wherever possible. A raw union bound over the thirteen maximal classes first gives lower bounds for e_G. It becomes equality only after uniqueness or all overlaps have been proved. The proofs use direct normalizer incidences, characteristic subgroups, small literal models, and separate generated subgroup types to account for these overlaps. This distinction matters: the bare 581-class roster would not by itself justify all 136 numbers.

The numerical component atlas in the binding list includes the prime columns $11, 19, 31$, the $5/7$ and 15 arguments, the block and normal overgroup arguments, the complete columns $14, 16, 20, 28$, the remaining small pairs and the final projective order-six calculation. The latter joins the exact projective contribution outside involution centralizers ($68g$ and $200g$) to the separately proved complementary contributions ($2g$ and $46g$), giving $(4,6) = 43$ and $(6,6) = 165$ after subtraction and division by $2g$. The accepted exact ledger joins all 136 entries to these proofs and their reviewed input contracts.

Summing these entries gives 798644. Exactly seven entries vanish. Five are also excluded by the elementary finite or Euclidean triangle quotients for $(2,3,3), (2,3,4), (2,3,5), (2,3,6), (2,4,4)$; the last two are the attributed $(3,7)$ exclusion and the separately proved $(3,12)$ proper-cover exclusion. Orders one or two cannot supply an omitted generating type: an order-one generator gives a cyclic group; if $R^2 = 1$ and $(RS)^2 = 1$, R inverts S, so the group is dihedral. This proves the complete rotation-type envelope in Theorem 1.1(1).

$p\backslash q$	3	4	5	6	7	8	10	11
3	0	0	0	0	0	10	7	37
4		0	18	43	102	284	285	503
5			26	98	150	470	365	718
6				165	354	1122	953	1776
7					648	1848	1506	2687
8						5424	4460	8096
10							3613	6532
11								11839

Table 1: Exact ordered A-orbit counts, first half of the triangular table.

$p\backslash q$	12	14	15	16	19	20	28	31
3	0	10	6	68	57	44	20	25
4	120	166	234	1292	846	554	370	359
5	211	290	340	1966	1242	800	560	502
6	474	597	874	4700	2943	1948	1370	1268
7	815	1054	1284	7448	4725	2916	2214	1995
8	2370	3056	3960	22040	13926	8880	6276	5752
10	1969	2526	3072	17822	11262	7224	4992	4632
11	3583	4601	5814	32488	20493	13094	9202	8325
12	1072	1391	1710	9764	6126	3984	2796	2577
14		1796	2330	12504	7899	5024	3616	3266
15			2834	15808	10020	6424	4496	4062
16				88784	56052	35644	25316	23048
19					35442	22572	15978	14553
20						14238	10292	9118
28							7246	6658
31								5999

Table 2: Exact ordered A-orbit counts, second half of the triangular table.

4 Inverters and string triples

Suppose (R, S) generates the centreless group G and an automorphism b inverts both. Two such automorphisms have quotient fixing R, S, so b is unique. Its square fixes both, and $b \neq 1$ because $p, q > 2$; therefore b is an involution. Set

$$a = Rb, \qquad c = bS.$$

Then a, b, c are nonidentity involutions, $ab = R$, $bc = S$ and $ac = RS$. Thus a, c commute and are distinct. Conversely any such triple with $\langle ab, bc\rangle = G$ supplies the simultaneous inverter b. The construction is equivariant under A. Since G has index two in A, all three involutions have the same parity. Inner triples generate G; all-outer triples generate A.

Here is the intersection-property argument, included to make the polyhedron inference explicit. The cyclic groups $\langle R\rangle$ and $\langle S\rangle$ intersect trivially, since an element of their intersection commutes with both generators and belongs to $Z(G)$. The adjacent dihedral groups are $D_0 = \langle R, b\rangle$ and $D_1 = \langle S, b\rangle$. If their intersection were larger than $\langle b\rangle$, comparison of forms $R^i b^\epsilon = S^j b^\delta$ would force a mixed-parity equality $R^i = S^j b$. For outer b this contradicts membership in G. For inner b, this common element is a reflection in D_1: it centralizes R and inverts S. Hence both $(RS)^2 = (RS^{-1})^2 = 1$. These relations make $\langle R^2, S^2\rangle$ abelian and normal, with quotient elementary abelian of order at most four: R inverts S^2, and S inverts R^2. They would make G solvable, a contradiction. Thus $D_0 \cap D_1 = \langle b\rangle$. The other pair intersections with $\langle a, c\rangle \cong C_2^2$ are respectively $\langle a\rangle$ and $\langle c\rangle$; otherwise the first equality would fail. These are the full rank-three intersection property.

Conversely, an inner string triple generating the simple group has rotation subgroup normal of index at most two; simplicity forces it to be G. For an all-outer triple generating A, the rotation subgroup lies in G and has index at most two in A, so again equals G. These observations recover the source's map/polyhedron assertions under the specified classical interpretation.

Reversal sends (a, b, c) to (c, b, a) and rotations to (S^{-1}, R^{-1}). It preserves inner/outer parity. It is not an additional identification in Tables 1–2. If d_p is the number of reversal-fixed inner orbits at (p, p), the number of inner strings after also identifying reversal is

$$\sum_{p<q} n_{\mathrm{in}}(p, q) + \sum_{p} \frac{n_{\mathrm{in}}(p, p) + d_p}{2}.$$

Thus neither the source's triangular total nor a type count determines the separate L026 rank-three total.

5 Chirality and exact small splits

There is one A-class of inner involutions and one of outer involutions. For a representative a_i, $i = 0, 1$, let $\kappa_m(i)$ count same-parity involutions b with $|a_i b| = m$, and put $h_i = |C_A(a_i)|$. Counting triples before imposing the second adjacent order or generation bounds reflexible orbits by

$$n_i(p, q) \leq \left\lfloor \frac{\kappa_2(i)\min(\kappa_p(i), \kappa_q(i))}{h_i} \right\rfloor.$$

The proof counts the choices of first involution, its partner and the commuting endpoint, then divides by the free action on generating triples. Reversal gives the minimum. Here $h_0 = 322560$, $h_1 = 351120$; all κ values are independently acquired class-product counts in the named table `ON.2`.

Subtracting both bounds from an acquired generating lower bound proves 100 positive chiral types. The separately acquired literal certificates prove another 23. Complete small-row arguments and the complete $(6, 6)$ split supply the remaining five. Their exact union is $T \setminus \{(3, 15)\}$,

of size 128. A certificate here proves a positive marking; an unfinished operation schedule is never used as an exhaustion certificate. The complete exceptional split is

$$n(3,15) = 6 = 5 \text{ inner} + 1 \text{ outer} + 0 \text{ chiral}.$$

The minimum-order row calculations give the following disjoint counts. For order five the independently checked order-four cross-check is excluded from this minimum-order sum.

Minimum order	All	Inner	Outer	Chiral
3	284	39	15	230
4	5176	114	156	4906
5	7738	188	210	7340

The individually asserted small splits are

Type	All	Inner	Outer	Chiral
$(3,10)$	7	0	1	6
$(4,5)$	18	0	2	16
$(4,6)$	43	0	1	42
$(5,5)$	26	2	2	22
$(6,6)$	165	8	1	156

These exact counts also settle the three isolated inner exclusions used below. The accepted minimum-order-five row supplies an inner $(5,7)$ marking. The diagonal transfer in Section 7 supplies the separately asserted inner $(7,7)$ marking.

6 Complete negative type envelopes

The acquired inverse-class and fusion calculation gives two G-classes of order 20 and two of order 31, with no inner inverter. The two order-eight and four order-sixteen classes have no outer inverter. These are necessary one-element conditions; no implication from separate invertibility of two generators to a common inverter is used. Together with the exact small zeros, every inner type lies in I.

There is one additional outer exclusion. If outer involutions a, b satisfy $|ab| = 6$, then $z = (ab)^3$ is an inner involution centralizing a, b. Normalize z and work in its acquired full centralizer of order 322560. The outer involutions there form one class of size 1344. For its representative, exactly 20 outer partners satisfy the normalized order-six relation; they form one complete orbit with stabilizer order 12. Transporting the first member to the fixed outer involution t puts the third member in the complete 1463-element domain from J_1. The acquired full stabilizer partitions that domain into 141 orbits, whose sizes sum to 1463. Their exact adjacent-product orders are

$$\{2, 3, 4, 5, 6, 7, 10, 11, 14, 15, 19, 20, 28, 31\};$$

order 12 never occurs. The order is computed on every orbit before any generation filter. Thus no outer triple of type $(6, 12)$ exists, and reversal excludes the other orientation. Every outer type therefore lies in J. This contradicts the particular outer existence assertion on source PDF page 9 without excluding inner $(6, 12)$ or changing the total pair count for that type.

The upper envelopes have sizes $|I| = 95$ and $|J| = 97$. Their proof does not require a complete census of all inversion orbits. Once a positive certificate or a proved positive count exists at every member of each envelope, equality follows. The exact positive union now equals these two sets; its proof inputs are described next.

7 Positive spectra and their final certificates

The positive union combines complete low-order rows, literal generating witnesses, algebraic operations on accepted witnesses, and two counting proofs. Their roles are different but all prove the same existential predicate for one type and parity. No expected source count is an input to a generation test.

For unit operations, a generating string (a, b, c) with $[a, c] = 1$ permits replacing the middle reflection by $a(ab)^e$ when $\gcd(e, |ab|) = 1$. The powers recover ab, hence recover b and the whole reflection subgroup. In the inner case that group is perfect, so its even-word subgroup cannot be proper of index two; in the outer case the even words generate the named index-two socle. Full word equalities, orders and parity are checked for each submitted result. Classical Petrie replacement $(a, b, c) \mapsto (ac, b, c)$ is used only on inner strings: the generated group is unchanged, and perfectness again forces the new rotations to generate it. Its attribution to the classical Petrie operation is retained. The argument is not applied to mixed parity outputs of an outer string.

The diagonal transfer uses an accepted string (x, a, b) with $|xa| = 4$ and $[x, b] = 1$. Put $c = a^x$. Then a, c are distinct commuting involutions, and $K = \langle a, b, c\rangle$ is normalized by x; it has index at most two in the original reflection group. For an inner string, perfectness gives $K = G$. For an all-outer string, $A' = G \leq K$ and $a \notin G$ give $K = A$. Thus (a, b, a^x) is a new generating string with equal adjacent orders $|ab|$. The already proved intersection property applies. This transfers the positive order-four rows to diagonal types, including the required inner $(7, 7)$ and $(10, 10)$ and four outer diagonal types. Subsequent fixed word calculations admit a positive type only after independent literal reconstruction.

The counting proofs for inner $(6, 10)$ and $(6, 28)$ do not infer generation merely from a free triple. For a selected triple with trivial A-stabilizer, the centralizer C of the rotation pair has $C \cap C_A(b) = 1$. Conjugation by the inner inverter b is consequently fixed-point-free on C, and the map $u \mapsto u^{-1}u^b$ is bijective. Its image is inverted by b, so C is abelian of odd order. It lies in G because $[A : G] = 2$. The acquired element centralizers of orders 72 and 20 (respectively 28) then force $C = 1$. Hence a free triple maps injectively to a pair before any generation assumption. Only after this multiplicity issue is settled are the raw maximal-overgroup bounds subtracted. The component proofs give at least 12 inner $(6, 10)$ orbits and at least two inner $(6, 28)$ orbits; they do not identify a particular generating representative or claim an exact count.

The last three complete pair-family acquisitions establish outer $(10, 12)$ with 12 ordered orbits, inner $(7, 14)$ with four, and inner $(7, 10)$ with three. For the final remaining positive type, an acquired inner $(4, 11)$ string (x, y, z) supplies a fixed first-order-eleven pair. Write $s = yz$, $z = x_3^w$ and use the acquired transport $x_3^{h_3} = x$. With right conjugation $u^v = v^{-1}uv$,

$$y^{s^6} = z, \qquad h = s^6 w^{-1} h_3, \qquad y^h = x.$$

The submitted third involution is the class-two representative conjugated by the signed word $(1, 2, -1, -1, -1, 2, -1, -1, 2)$ in the acquired centralizer generators. Independent reconstruction checks the complete pair, transport and rotation hashes. Direct full-array prime powers prove the exact orders $11, 7, 2$ and the inversion equalities. A separately recomputed complete point orbit has size 122760. Every proper subgroup is contained in a maximal subgroup, whereas none of the thirteen maximal orders is divisible by 122760. Orbit–stabilizer therefore proves generation. This acquires one inner $(7, 11)$ marking. It reuses the named model and the generation predicate but checks the new orders by a different direct-power path. It does not replay the 45 earlier negative candidates or assert a complete count at this type.

The immutable positive union, including this final certificate, equals I and J. Combining it with Section 6 proves Theorem 1.1(3)–(4). An interrupted word schedule was never used to assert absence. The completed type spectra likewise assert no uncomputed total number of reflexible markings.

8 The source's constructive algorithm

Fix an order $p > 2$. Choose one representative s of every G-class of order p, then one involution τ from every $\langle s \rangle$-orbit on the involutions of G. Test

$$(r, s) = (\tau s^{-1}, s), \qquad |r| \geq p, \qquad \langle r, s \rangle = G,$$

and retain one representative under the full automorphism action. Every required pair can first have its second entry conjugated to a chosen s, and then its product involution moved to τ by a power of s without changing s. This proves coverage. The final full automorphism comparison removes duplicates. Using the smaller cyclic group rather than the whole centralizer affects redundancy, not completeness; a cyclic orbit is not silently called a full centralizer orbit.

The conjugation action on involutions is the action on right cosets of $C_G(z)$ via $C_G(z)g \mapsto z^g$. It is faithful by simplicity and has degree $g/161280 = 2857239$. Its points can retain their involution labels; no unproved extraction from an abstract stabilizer is needed. The complete roster also bounds every nonidentity centralizer by 161280: an element centralizer lies inside the normalizer of its cyclic subgroup, and the maximum normalizer among all roster rows of that order is an upper bound. This uses all relevant rows rather than guessing which rows are cyclic.

Stopping when a known count is reached is valid only after checking that every retained marking passes the tests and no two are equivalent. An inclusion of finite sets with equal cardinality is equality. The stopping count for $(20, 31)$ must be 9118, and the aggregate must be 798644. No runtime bound follows from this completeness proof. The historical Magma runs and times reported in the source are not replayed here.

9 Source corrections and limits

Statement	Printed source	Acquired replacement
Table 1, $(20, 31)$	7246	9118
Aggregate triangular count	796772	798644
$(3, 15)$ reflexible split	4 inner, 2 outer	5 inner, 1 outer; no chiral
Outer type spectrum	98; includes $(6, 12)$	97; excludes $(6, 12)$
Adjacent $(3, 12)$ prose	Both zero and positive	Zero; independent proper-cover proof

The accepted independent proof of the $(3, 12)$ zero does not validate the inadequate cited sample-pair argument. The source's complete table and its mathematical claims are treated individually, including claims that require correction. Historical computations remain historical; the source's suggested future investigation of exponents and self-duality is context, not an assertion of a completed census.

The rank-three reversal results derived along the way (114 inner and 156 all-outer reversal-fixed markings under the stated A action) do not settle L026's separate rank-three total or its higher-rank claims. Likewise this treatment adds no Lean formalization, general conjecture resolution or theorem-priority claim. Whole-source admission and public registration are review actions separate from this mathematical assembly.

10 Evidence, reproduction and source concordance

The canonical text is this TeX file, its generated exact-table and source-concordance inputs, and the companion structured binding list. The assembly checker performs only finite integer/set and metadata joins: it checks the 136 entries, unique source correction, aggregate, small splits, 95/97 envelope equality and all 44 clause identifiers against their exact acquired inputs. It does not relaunch a permutation census, subgroup computation or witness search. The historical computational proofs retain their own independently acquired receipts and input scopes.

The new binding list uses paths relative to the retained study root. Publication must include the transitive canonical proof/code/input and accepted-certificate closure, with an explicit resolver for absolute paths inside unchanged historical records. A local synthesis check by itself does not establish that those private files are already protected by a public clone. Build scratch files are not mathematical prerequisites of this manuscript. Historical records that themselves bind such files must remain unaltered; a newly reviewed portable interface must distinguish their audit history from the canonical reproduction inputs rather than silently deleting a pinned premise.

Clause	PDF pp.	Disposition	Section
T1.1.1	3,7	recovered from accepted components	1
T1.1.2	1,3,7	corrected complete	1
T1.1.3	3,8	recovered complete chiral spectrum with corrected exception	1
T1.1.4	3,8	recovered complete 95 inner types	1
T1.1.5	3,8,9	corrected complete 97 outer types	1
definitions	2,3,4,7	recovered or explicit classical premise	4
T2.1	4	attributed borrowed theorem	3
C2.2	5	recovered	3
L2.3	5	recovered	3
equation2.4	5	recovered	3
algorithm3	5,6	recovered	3
mobius	6	recovered general argument	3
ONpremises	7,9	explicit borrowed inputs	2
p2excluded	7	elementary corollary	3
table1	11	complete with one correction	3
source_prior_results	7	attributed context	9
AlKadhi312	7	zero reproved and prose corrected	9
L5.1	8	corrected complete chiral spectrum and exception	5
L5.2	8	recovered	6
L5.3	8	recovered	6
split_3_10	8	recovered from complete row	5
split_4_5	8,9	recovered from complete row	5
split_4_6	8	recovered from complete row	5
split_5_5	8,9	recovered from complete row	5
inner_witness_5_7	8	recovered from complete row	7
inner_witness_7_7	8	recovered diagonal transfer	7
outer_witness_6_12	9	refuted by complete accepted outer involution domain	6
algorithm6	9	recovered with explicit completeness and corrected stopping total	8
polyhedra	9	recovered with historical attribution	4
row_total_3	9	recovered arithmetic	5
row_chiral_3	9,10	recovered from complete row	5
row_inner_3	9,10	recovered from complete row	5
row_total_4	9	recovered arithmetic	5
row_chiral_4	9,10	recovered from complete row	5
row_inner_4	9,10	recovered from complete row	5
row_total_5	9	recovered arithmetic	5
row_chiral_5	9,10	recovered from complete row	5
row_inner_5	9,10	recovered from complete row	5
historical_timing	7,8,9,10	attributed historical observations	9
prevalence	10	qualitative context	9
outer_row_3	3,8,9	complete with correction	5
inner_row_3	3,8,9	complete with correction	5
complete_row_4	8,9	complete	5

Clause	PDF pp.	Disposition	Section
complete_row_5	8,9,10	complete	5

References

[1] T. Connor and D. Leemans, Algorithmic enumeration of regular maps, *Ars Math. Contemp.* 10 (2016), 211–222.

[2] T. Connor and D. Leemans, An atlas of subgroup lattices of finite almost simple groups, *Ars Math. Contemp.* 8 (2015), 259–266.

[3] G. Frobenius, Ueber Gruppencharaktere, *Berl. Ber.* (1896), 985–1021.

[4] P. Hall, The Eulerian functions of a group, *Q. J. Math., Oxford Ser.* 7 (1936), 134–151.

[5] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker and R. A. Wilson, *Atlas of Finite Groups*, Oxford University Press, 1985.

[6] T. Breuer, *The GAP Character Table Library*, retained installed tables and exact acquisition interfaces.

[7] A. J. Woldar, On Hurwitz generation and genus actions of sporadic groups, *Illinois J. Math.* 33 (1989), 416–437.